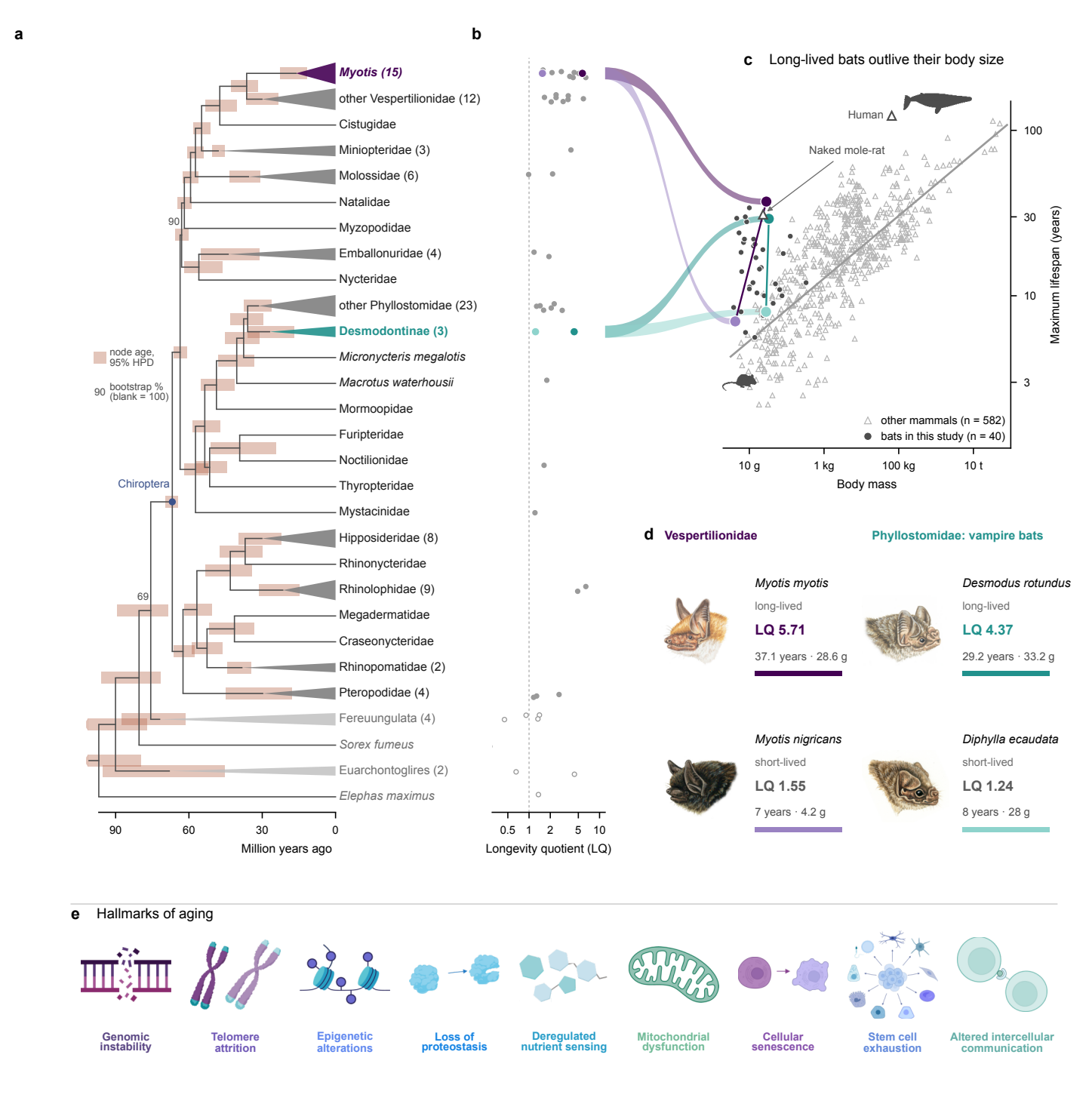

a
Myotis (15)
other Vespertilionidae (12)
Cistugidae
Miniopteridae (3)
Molossidae (6)
Natalidae
Myzopodidae
Emballonuridae (4)
Nycteridae
other Phyllostomidae (23)
Desmodontinae (3)
Micronycteris megalotis
Macrotus waterhousii
Mormoopidae
Furipteridae
Noctilionidae
Thyropteridae
Mystacinidae
Hipposideridae (8)
Rhinonycteridae
Rhinolophidae (9)
Megadermatidae
Craseonycteridae
Rhinopomatidae (2)
Pteropodidae (4)
Fereuungulata (4)
Sorex fumeus
Euarchontoglires (2)
Elephas maximus
Chiroptera
90
69
node age, 95% HPD
90 bootstrap % (blank = 100)
90 60 30 0
Million years ago
b
0.5 1 2 5 10
Longevity quotient (LQ)
c Long-lived bats outlive their body size
Human
Naked mole-rat
100
30
10
3
Maximum lifespan (years)
other mammals (n = 582)
bats in this study (n = 40)
10 g 1 kg 100 kg 10 t
Body mass
d
Vespertilionidae
Phyllostomidae: vampire bats
Myotis myotis
long-lived
LQ 5.71
37.1 years · 28.6 g
Desmodus rotundus
long-lived
LQ 4.37
29.2 years · 33.2 g
Myotis nigricans
short-lived
LQ 1.55
7 years · 4.2 g
Diphylla ecaudata
short-lived
LQ 1.24
8 years · 28 g
e Hallmarks of aging
Genomic instability
Telomere attrition
Epigenetic alterations
Loss of proteostasis
Deregulated nutrient sensing
Mitochondrial dysfunction
Cellular senescence
Stem cell exhaustion
Altered intercellular communication